

Scenario	kNN	Mahalanobis	CKS	Ridge	Feature Selection
Baseline & Injury, No Treatment Baseline & Injury, Single Treatment	Yes, normalized	Yes	Yes	Yes, no interaction	Feature selection comparing injury to baseline.
Baseline Only, No Treatment Baseline Only, Single Treatment Injury Only, Not Treatment Injury Only, Single Treatment	Yes, raw	No	No	No	No
Baseline Only, Multiple Treatments Injury Only, Multiple Treatments	Yes, raw	No	No	No	Feature selection comparing treatment effects
Baseline and Injury, Multiple Treatments	Yes, normalized	Yes	Yes	Yes	Feature selection comparing injury to baseline. Feature selection comparing treatment effects

Clarification of scenarios:

- 1. Baseline and Injury, No Treatment:** Data consists of uninjured and injured rodents. Baseline-injury axis can be computed, but there are no multiple treatment groups to compare.
- 2. Baseline and Injury, Single Treatment:** Data consists of uninjured and injured rodents. Baseline-injury axis can be computed, but there are no multiple treatment groups to compare.

3. **Baseline Only, No Treatment:** Data consists of only uninjured animals, so baseline-injury axis cannot be computed. No treatment groups to compare.
4. **Baseline Only, Single Treatment:** Data consists of only uninjured animals, so baseline-injury axis cannot be computed. No treatment groups to compare.
5. **Injured Only, No Treatment:** Data consists of only injured animals, so baseline-injury axis cannot be computed. No treatment groups to compare.
6. **Injured Only, Single Treatment:** Data consists of only injured animals, so baseline-injury axis cannot be computed. No treatment groups to compare.
7. **Baseline Only, Multiple Treatment:** Data consists of only uninjured animals, so baseline-injury axis cannot be computed. But multiple treatment groups exist to compare.
8. **Injured Only, Multiple Treatment:** Data consists of only injured animals, so baseline-injury axis cannot be computed. But multiple treatment groups exist to compare.
9. **Baseline and Injured, Multiple Treatment:** Data consists of baseline and injured animals, so baseline-injury axis can be computed. Multiple treatment groups exist to compare.

Note:

- When there is no injury it cant compare so the change to the logic would be to have the raw values to report for kNN, meaning no scaling. So when the injured data is missing, it should skip the scaling step and feed raw values directly into kNN/Mahalanobis instead of standardized ones.